



State of Washington
DEPARTMENT OF FISH AND WILDLIFE

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June 30, 2025

The Honorable Mike Chapman
Chair, Senate Agriculture and
Natural Resources
402 Legislative Building
Post Office Box 40424
Olympia, WA 998504

The Honorable Kristine Reeves
Chair, House Agriculture and
Natural Resources
132F Legislative Building
Post Office Box 40600
Olympia, WA 98504

Dear Chairs Chapman and Reeves:

I am pleased to provide you with the Avian Salmon Predation Working Group (ASPWG)'s report to the legislature. The report was prepared by the ASPWG as directed in Substitute House Bill 2293 (2024), "Studying the effects of avian predation on salmon."

The ASPWG was tasked with identifying all avian species that contribute to the predation of juvenile salmon at a population level, determining whether such species are adversely impacting the recovery of any threatened or endangered salmon species, and identifying remedies to predation. The report compiles useful data and existing knowledge about avian salmon predation in Washington and makes recommendations for how the legislature can support the Washington Department of Fish and Wildlife (WDFW) on this complex situation.

In summary, the ASPWG recommends:

1. Fund research to examine geographically specific anecdotes of salmon predation by birds, particularly Puget Sound rivers with at-risk salmon stocks.
2. Invest in WDFW's capacity to coordinate with federal and state governments, tribes, and other regional entities to address avian predation issues in the Columbia River System.
3. Invest in Department-led work focused on closing data gaps and learning more about the shared needs of birds and salmon.

The legislature appropriated \$222,000 in the 2024 supplemental operating budget to WDFW for fiscal year 2025 to implement SHB 2293. The funding was used to convene the working group, pay for meeting facilitation, and pay staff and a contractor to help the ASPWG draft the report. SHB 2293 prescribed the sectors to be represented as participants in the ASPWG. The ASPWG met once in 2024 and four times in 2025. Meetings were either fully virtual or "hybrid" with an in-person opportunity in Olympia. All meetings were open to the public.

I want to express my appreciation to the ASPWG for their engagement and contributions to the report. Managing species to recover other species engenders strong opinions and perspectives;

Sen. Chapman and Rep. Reeves

June 30, 2025

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WDFW appreciated the ASPWG members' commitment to constructive engagement and active, careful listening to each other as they deliberated on the science and management principles and worked to develop their report.

If you have any questions about this report, please reach out to our legislative director, Melena Thompson, at 564-791-2755.

Sincerely,

A handwritten signature in blue ink, appearing to read "Kelly Susewind", is written over a light blue horizontal line.

Kelly Susewind
Director

cc: Owen Rowe, Natural Resources Policy Advisor to Gov. Ferguson
Washington Fish and Wildlife Commission
Nate Pamplin, Director of External Affairs

Avian Salmon Predation

Working Group

Report to the Washington State Legislature



Avian Salmon Predation Working Group

2025 Report to the Washington State Legislature

Authors

Jessica Stocking (WDFW) and Shelby Thomas (Ross Strategic) relied on ASPWG meeting notes to draft the report on behalf of the ASPWG. All ASPWG members were invited to review and revise the draft report.

Report acknowledgements

Special thanks to Alison Anholt, James Lawonn, Jennifer Urmston, Jeremy Cram, and Todd Hass for their additional efforts to draft and review specific sections of the report particularly relevant to their areas of expertise.

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Cover image of Caspian tern; Original 19th Century engraving by Benjamin Fawcett.

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Common acronyms

ASPWG or WG – Avian Salmon Predation Working Group or Working Group

CRS – Columbia River System

DPS – Distinct Population Segment

ESA – Endangered Species Act

ESU – Evolutionarily Significant Unit

MBTA – Migratory Bird Treaty Act

NMFS – National Marine Fisheries Service (NOAA Fisheries)

NOAA – National Oceanic and Atmospheric Administration

USACE – United States Army Corps of Engineers

USFWS – United States Fish and Wildlife Service

WDFW – Washington Department of Fish and Wildlife

Acknowledging the Indigenous People of the Pacific Northwest

Since time immemorial, Indigenous People have lived in the Pacific Northwest and hunted, fished, and gathered natural resources, traditional foods, and medicinal plants to support their diverse cultures. They were the original occupants and stewards of this land that all Washingtonians enjoy today.

The very survival of the Pacific Northwest Tribes is a testament to resiliency of what they have endured and continue to endure throughout generations on this landscape. Through many historical encounters of massacre, renunciation of religious freedom, systemic racism, cultural assimilation of Native children through institutional residential schools, and the fight for their inherent rights and liberties, they have prevailed. Throughout this painful history brought by colonization, abrogated treaties, infringement of civil rights, and the salmon protests of the 1960s, the Northwest Tribes and the Washington Department of Fish and Wildlife (WDFW) have founded a commitment of respect, unity, and alliance informed by the realities of the past.

Today, Tribal governments and WDFW work collaboratively to conserve and manage aquatic and terrestrial resources statewide and practice sound science to guide management decisions. The Tribes and WDFW work together to ensure the sustainability of fish, wildlife, ecosystems, and culture for the next seven generations and beyond.

Executive summary

In response to the Washington State Legislature’s 2024 [Substitute House Bill 2293](#) (SHB 2293), the Washington Department of Fish and Wildlife (WDFW, the Department) convened a diverse [Avian Salmon Predation Working Group](#) (ASPWG, or Working Group) to (1) identify all avian species that contribute to predation of juvenile salmon at a population level; (2) determine whether such species are adversely impacting the recovery of any threatened or endangered salmon species; and (3) identify remedies.

Although avian predation of fish is one part of a broad, complex natural resource issue, the ASPWG’s work was—by necessity and request—narrowly focused and allowed little space to explore complicated context and knowledge gaps. In responding to the Legislature’s requests, the Working Group discussed and acknowledged its limitations, which include the following topics:

- Predator-prey relationships
- Interconnected threats to avian and fish species
- Heavily modified natural environments
- Measuring impacts and effects
- Additive versus compensatory mortality
- Avian diet information
- Asymmetrical regional data and knowledge
- Lack of clear management goals

ASPWG high-level findings

- There are four primary bird species or species groups that have life history characteristics that may result in negative impacts to juvenile salmonids at a population level: American white pelicans, Caspian terns, Double-crested cormorants, and gulls.
- Determining whether instances of avian predation impact juvenile salmon survival must be done on a case-by-case basis; it is impossible to make a general statement such as “Bird species [X] impacts threatened or endangered salmonid species [X].” The ASPWG endorses the Pacific Flyway Council’s guidance on determining migratory bird predation impacts on fish resources, which states, “Responses to perceived avian predation issues [should be] based on sound science.”
- The ASPWG developed two products to respond to the Legislature’s request that they “identify remedies:”
 - Intervention Principles that guide the development of robust, long-term adaptive management plans for reducing avian predation, increasing juvenile salmonid survival, and building resiliency into salmon populations.
 - An inventory of actions to dissuade avian predation that have been implemented or are being implemented in the region.

ASPWG recommendations

The ASPWG recommends the Legislature consider the following to navigate the complex topic of avian predation on salmonids and overcome many of the knowledge gaps and limitations identified above:

1. Fund research to examine geographically specific anecdotes of salmon predation by birds, particularly Puget Sound rivers with at-risk salmon stocks and that are cited as places where avian salmon predation negatively impacts juvenile salmonid survival, including but not limited to the Stillaguamish River and the Nisqually River Delta.
2. Invest in WDFW's capacity to coordinate with federal and state governments, tribes, and other regional entities to address avian predation issues in the Columbia River System.
3. Invest in Department-led work focused on closing data gaps and learning more about the shared needs of birds and salmon. This work can make critical connections between ecosystem recovery actions that benefit both species, increase the Department's ability to partner in avian predation policy and science, and promote coordination with tribes and other partners. For example:
 - a. Update Washington seabird colony inventory and diet database.
 - b. Research piscivorous bird species diet and foraging dynamics in Washington state.

Report purpose

In 2024, the Washington State Legislature passed [Substitute House Bill 2293](#) (SHB 2293), which required the Washington Department of Fish and Wildlife to convene a diverse Avian Salmon Predation Working Group to produce a report for the Legislature by June 30, 2025, that responds to the following three requests:

1. Identify all avian species that contribute to predation of juvenile salmon at a population level.
2. Determine whether such species are adversely impacting the recovery of any threatened or endangered salmon species.
3. Identify remedies.

This report fulfills the Department and ASPWG requirements as outlined in SHB 2293. The report includes background and context critical to understanding the complexity behind each of the Legislature's requests. The report also includes the ASPWG's high-level responses and important considerations to accompany each response, a list of ASPWG-recommended intervention principles, and recommendations.

Please see Appendix A, "ASPWG membership" and Appendix B, "ASPWG process overview" to learn more about the Working Group's membership and meetings.

Background and context

Predation is a natural, dynamic, and complex ecosystem process. Both predator and prey species are intrinsically valuable components of naturally functioning ecosystems and contribute to biodiversity in Washington. Many species of birds in Washington eat fish, especially small and/or schooling fish, including juvenile salmonids. Abundance and distribution of birds and fish, as well as the predation process, have been altered by human activities and structures. In some cases, there have been attempts to characterize and control predation to benefit salmonids, resulting in varied outcomes and conclusions.

Although avian predation of fish is one part of a broad, complex natural resource issue, this report is, by necessity and request, narrowly focused, allowing little space to explore the complicated context. The limitations faced by the ASPWG are highlighted in this report's "Complexities and knowledge gaps" section; the reader should note that these limitations are interconnected and, when taken together, illustrate the difficulty in responding directly to the Legislature's requests.

Brief history of avian predation management in Washington

Lethal or non-lethal predator management can be a viable management tool to increase survival of prey populations targeted for recovery. This type of management action is typically pursued when the ecosystem or ecosystem processes are already heavily influenced by humans. It may be possible to

increase prey population through predator management when addressing other threats to the prey population is not feasible or desirable, habitats are highly altered, and predation is documented as having a disproportionately large impact on a prey species, typically during a vulnerable life history stage, compared to other sources of mortality.

Based on many years of research along the Columbia River (Roby et al. 2021) and driven by Federal Columbia River Power System Biological Opinions issued by the National Marine Fisheries Service, three management plans were developed by the U.S. Army Corps of Engineers, the U.S. Fish and Wildlife Service (USFWS), and the U.S. Bureau of Reclamation to attempt to reduce the impacts of predation by Caspian terns and double-crested cormorants on juvenile salmonids listed under the Endangered Species Act (ESA). These plans prescribe specific actions for breeding colony size reductions, including planting vegetation, hazing, egg take, and lethally removing individuals to reach target nesting populations to reduce impacts on threatened and endangered salmonids. Two of the plans addressed Caspian terns and double-crested cormorants in the Columbia River estuary, and one addressed terns nesting at inland islands. Some conclusions drawn from actions conducted under the three management plans include:

- Predation by Caspian terns on three islands specified in the plans has decreased with targeted management efforts.
- Double-crested cormorants were successfully reduced on East Sand Island in the Columbia River estuary, which caused a shift to use of areas in the upper estuary and increased local predation at the Astoria-Megler Bridge (Oregon Department of Fish and Wildlife, unpublished data). Adaptive management continues to be necessary to achieve reduced predation impacts by cormorants.
- Management of Caspian terns since 2008 has reduced the colony on East Sand Island to below the target colony size in the plan, and the colony has experienced complete nesting failure in six out of the last nine years (2016-2024; Evans et al. 2025). Additionally, the Pacific Flyway population has decreased by more than 50% since management began (in 2008), triggering an adaptive management threshold for reducing the decline.

Avian management regulatory framework

The ASPWG's considerations and responses to SHB 2293 are bound by current federal and state regulations related to avian management, specifically the Migratory Bird Treaty Act and tribal and treaty rights.

Migratory Bird Treaty Act

The [Migratory Bird Treaty Act](#) (MBTA) is a federal law enacted in 1918 to protect migratory birds and ensure the sustainability of bird populations in the United States. It implements international treaties with Canada, Mexico, Japan, and Russia, making it unlawful to pursue, take, capture, kill, or possess any

migratory bird, part, nest, or egg without authorization, i.e., a permit, from the USFWS [Migratory Bird Permit Program](#).¹

USFWS can [authorize exceptions](#) to the MBTA for several reasons. The USFWS's most common authorizations are depredation permits; other permits exist for scientific collection, rehabilitation, education, and others. Two permit types are germane to the ASPWG's work: The Depredation Permit and the Special Double-crested Cormorant Permit. Depredation permits are often issued when there are demonstrated instances of birds impacting human health, safety, or property; threatened or endangered species; or causing losses to agriculture/aquaculture. The Special Double-crested Cormorant Permit can be issued to state or tribal fish and wildlife agencies and authorizes take of cormorants to reduce or prevent fish depredation at state- or tribal-owned aquaculture facilities (e.g., hatcheries); impacts to human health and safety; impacts to threatened and endangered species; and damage to state or tribal property and assets.

Tribal and treaty rights

In Washington, treaty tribes and the state co-manage salmon, steelhead, and other fisheries. This means they jointly determine fishing seasons and hatchery production objectives, and collaborate on recovery efforts, harvest levels, and enforcement. The tribes and the state use a unique government-to-government process that is designed to honor the tribes' rights reserved in several treaties signed with the federal government in the 1850s, and to elevate the tribal knowledge and traditional practices that are critical in understanding and managing salmon populations and sustainable fisheries.²

Salmon are central to tribes' cultures, identities, and businesses; salmon recovery is imperative to their way of life. Although treaties do not explicitly reference avian management, they guarantee the presence of and access to fish populations that are in severe decline due to a combination of factors such as degraded and heavily modified habitat, rising stream temperatures, reduced fresh and marine water quality, and increased predation. Agencies pursuing avian management plans should consult with tribes when setting population management objectives for birds and fish.

¹ The term "take" encompasses actions such as touching, trapping, relocating active nests, and flushing in a manner that leads to nest failure. Hazing birds does NOT require authorization and can include activities such as auditory dissuasion, falconry abatement, habitat modification to disturb nesting behaviors, use of propane cannons, use of green lasers, and flushing birds with dogs. This is slightly different from flushing in a manner that leads to nest failure.

² Treaties include Treaty of Medicine Creek (1854), Yakama Nation Treaty (1855), Treaty of Walla Walla (1855), Treaty of Neah Bay (1855), Treaty of Point Elliot (1855), Treaty of Point No Point (1855), and Quinault Treaty (1856). Treaty rights were further reinforced by two court cases: *US v. Washington* (aka Boldt Decision) and *US v. Oregon*.

The exclusive right of taking fish in all the streams...is further secured to said [Tribes], as also the right of taking fish at all usual and accustomed places...*Article 3 of the Yakima Nation Treaty of 1855*

State of current knowledge

Piscivorous birds in Washington

Appendix C, “Table of Washington piscivorous birds,” details the piscivorous bird species commonly found in Washington and relevant factors that can be used to determine their potential impact to listed salmonids, including prey selection, avian foraging type, and seasonal occurrence. Impacts to salmonids vary based on the specific phenology of breeding activities which have different energetic demands (timing of arrival, incubation, and chick-rearing), and on species abundance, composition, and location.

The body of work around avian predation and its management in the Columbia River System (CRS) points to focal species that are potentially impacting salmon recovery: double-crested cormorants, Caspian terns, American white pelicans; and, to a lesser extent, California and ring-billed gulls. In a few well-studied locations, there is information about these birds’ impacts on specific salmon runs, but those are the exceptions. Here we provide additional information on the presence and diet of the species in these four guilds in Washington.

Cormorants

Three cormorant species nest in colonies in Washington: pelagic, Brandt’s, and double-crested. While all three species eat fish as a primary part of their diet, pelagic and Brandt’s cormorants are restricted to marine or estuarine environments, where they eat a variety of fish and invertebrate species. Double-crested cormorants are adaptable to most water bodies in Washington and can utilize a variety of natural and manmade structures (e.g., channel markers, bridges, utility transmission towers). As such, their breeding colonies can be proximate to out-migrating smolt in freshwater locations. No known studies have found evidence that cormorants eat salmonids in the Puget Sound or Salish Sea, but multiple studies have found that double-crested cormorants eat salmonids in the Columbia River Basin (CRB; Roby et al. 2021; Appendix D). The western double-crested cormorant population has declined in recent years (Table 1).

Terns

Three species of tern regularly occur in Washington during the breeding season, but only two eat fish: Caspian and Forster’s terns. Washington’s small population of Forster’s terns nests in freshwater wetland colonies in Southeastern Washington. No information exists on their diet composition in Washington.

Caspian terns are well studied in the CRS. Traditionally, they nested on predator-free islands in freshwater lakes and rivers, but water shortages have forced them to abandon these habitats. The loss of historic habitat and new alternative habitats have led to a distributional shift northward and increase in Washington, but the Pacific Flyway population has declined by at least half since 2010 (Table 1). While the proportion of juvenile salmonids documented in the diet of adult and juvenile Caspian terns is highly variable and can vary across years (17-60%, Roby et al. 2005), the number and size of breeding colonies across the region have been relatively small compared to the Columbia River and thus seem unlikely to have broad effects.

Pelicans

Brown pelicans and American white pelicans are both present in Washington for part of the year. Brown pelicans are restricted to marine and estuarine habitats and eat a variety of fish and other prey items near the water surface, foraging by plunge-diving headfirst. Brown pelicans do not breed in Washington and are only present during the post-breeding season (May-November).

American white pelicans nest colonially on predator-free islands in fresh water, and forage primarily in groups, coordinating to concentrate prey into shallow waters. They have also been documented foraging at dam spillways solo or in groups. Recent nesting colonies include Badger Island and Miller Sands Spit in the Columbia River, although at the time of this writing, the Badger Island colony appears to be abandoned in 2025. Virtually all white pelicans migrate out of the state during the wintertime. There is some evidence that both the Flyway population and the Washington breeding colony have declined in recent years; they are listed as state sensitive. White pelicans eat a wide variety of fish and sometimes other prey items, including salmon and fish that consume salmon.

Gulls

Over ten species of gulls occur in Washington, but four species have widespread occurrence throughout the year: California gull, Western gull, ring-billed gull, and “Olympic” gull (a widespread marine gull which is a hybrid of Western and glaucous-winged gulls). California and ring-billed gulls are mostly freshwater dependent, and Western and Olympic are mostly marine dependent. These four species nest in colonies, sometimes alongside Caspian terns. Gulls are incredibly diverse in their diets, and unable to dive or swim underwater. Notably, gulls steal food from other fish-eating species such as terns and scavenge dead prey items. These complex behaviors make accurate estimates of direct predation difficult.

Standardized population assessments have been conducted in Washington and across the Pacific Flyway for double-crested cormorants, American white pelicans, and Caspian terns in collaboration with USFWS (Table 1).

Table 1. Pacific-flyway estimates of three piscivorous bird species of concern. Data provided by the Pacific Flyway Council.

Species	2014	2015	2016	2017	2018	2021	2022	2024
American white pelicans ¹	42,692	--	--	46,081	50,366	33,137	18,968 *	--
Double-crested cormorants ¹	73,437	72,304	71,585	43,074	59,571	49,698	--	**
Caspian terns ²	--	10,124	--	--	8,691	5,864	--	4,910

¹ individuals

² breeding pairs

* census incomplete; several known colonies were not surveyed in 2022

** surveys complete; awaiting analysis

Salmon in Washington

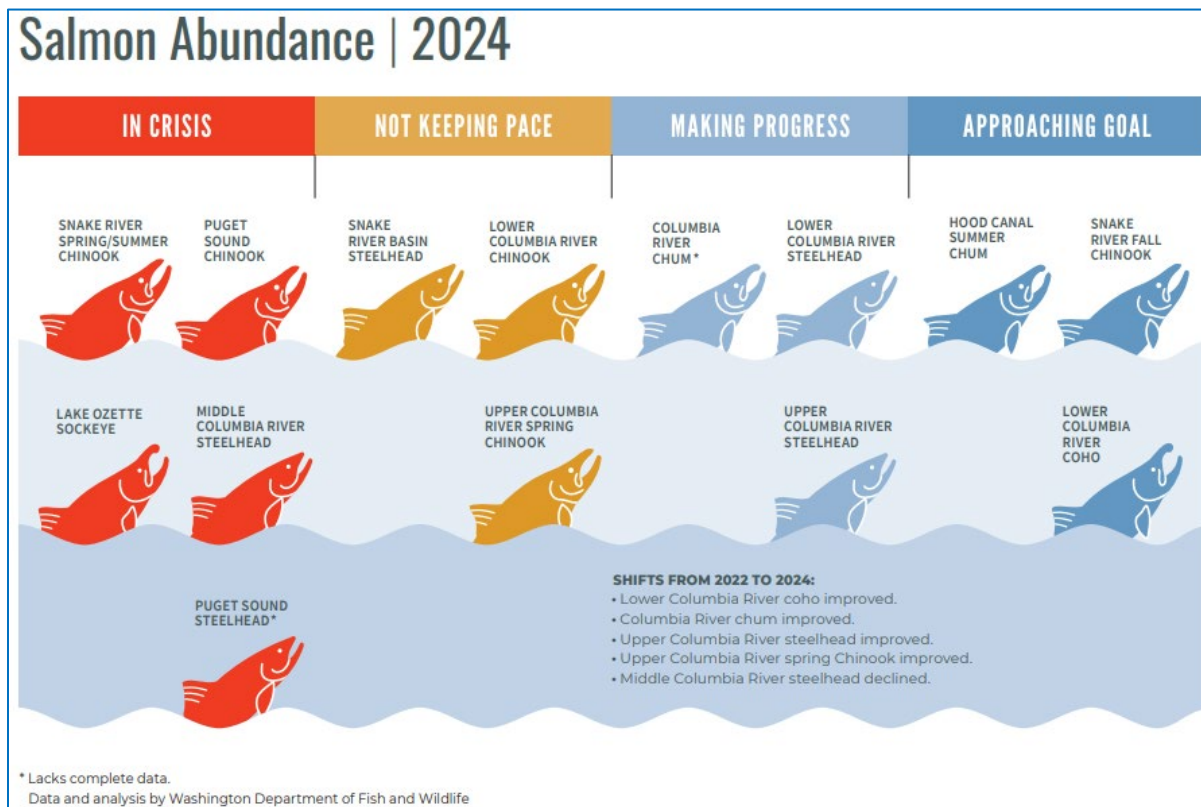
There are 14 Evolutionarily Significant Units (ESUs; applies to salmon) or Distinct Population Segments (DPSs; applies to steelhead) originating in Washington that are listed under the Endangered Species Act (Figure 1). Each of these population groups has an associated federal recovery plan that is approved by the National Oceanic and Atmospheric Administration (NOAA). Recovery plans generally detail the minimum viability criteria that would lead to delisting goals, meaning that the populations are no longer in danger of extinction, as well as listing known threats for each population. Satisfying the ESA is a critical, but narrow component of salmon recovery that is directed at avoiding extinction; more broadly, the objective is a more robust “healthy and harvestable” target that reflects the importance of salmon to Pacific Northwest cultures, economies, and ecosystems.

In addition to the federal recovery plans, Washington tracks and manages salmon recovery at the state level through the [Governor’s Salmon Recovery Office](#). At a smaller scale, salmon recovery in Washington is divided into regions, each with a salmon recovery board that prioritizes projects and funding and works with partners to implement recovery actions.

Most federal salmon recovery plans cite some number of the “4 Hs”—Habitat, Harvest, Hatchery, and Hydropower—as potential or demonstrated impediments to recovery. Similarly, most point toward native and non-native fish predation as a potential concern. Where avian predation appears in salmon recovery plans, details are often somewhat limited, and direction is provided in subsequent documents (see the three management plans for avian predation, referenced previously).

Figure 1. Salmon in Washington

Salmon abundance categories for the 14 ESA-listed ESUs/DPSs in Washington, excerpted from Washington’s [2024 State of Salmon in Watersheds executive summary](#). The Working Group recognizes the SOS work reflects only Washington populations within these ESUs/DPSs, and there may be reason to consider some in different abundance categories.



Avian salmon predation in Washington

While legacy habitat degradation is the primary challenge to salmon recovery in Washington, there are several other factors that must be evaluated and addressed to ensure viability of the most imperiled populations. These include climate change, water quality and quantity, impacts of hydropower operations, fish passage, food resources, harvest, hatcheries, and predation.

The degree to which each of the factors impacts salmon populations is difficult to quantify. A recent attempt to put predation in perspective of all identified threats to salmon recovery was conducted by the Columbia River Partnership Task Force (2020; Figure 2). Predation (in this case, by marine mammals, fish, and birds combined) typically accounts for a small portion of mortality relative to other threats, but there are exceptions. Appendix D, “Avian salmon predation in Washington” details locations of concern for listed salmon populations, citing avian predation statistics where reported.

The following three subsections provide key takeaways on avian salmon predation available to the ASPWG for different regions of the state, starting with the Columbia River system, which has the most mature body of work; the Puget Sound, which has a modest amount of information; and the coast, for

which there is little information. This asymmetrical set of information is explored further in the report's "Complexities and knowledge gaps" section.

Columbia River system

Most of what has been documented about avian predation on salmonids in Washington has occurred in the CRS, where there is a large body of work around predation and its management (Roby et al. 2021; Evans et al. 2024; Appendix D). Some lessons from this work include:

- Consumption of salmonid smolts by piscivorous birds is highly variable across bird species, fish population, and site; not all species/colonies threaten smolt survival in the CRS.
- Impacts to salmonids vary based on the specific phenology of breeding activities and the associated energetic demands (timing of colony arrival, incubation, and chick-rearing), as well as colony size, location, and species composition.
- Bird colonies in the Columbia River estuary, where there are a variety of available fish prey species, generally consume less salmon as a percentage of their total diet than bird colonies upriver (Roby et al. 2021).

Puget Sound

Compared to the CRS, effects of avian predation on juvenile salmonids have not been as well studied in the Salish Sea or associated riverine systems. The best synthesis on this topic assessed the state of evidence of salmonid predation by piscivorous marine birds and mammals in the region (Pearson et al. 2015). Pearson et al. (2015) used several criteria to identify potential salmonid and steelhead bird and mammal predators in the marine environment including: 1) the overlap in the timing of steelhead outmigration and bird use of the Salish Sea (e.g., migratory species), 2) the overlap in prey size with out-migrating salmon and steelhead, 3) evidence that the marine bird predator consumes juvenile salmon and, 4) evidence that the marine bird predator consumes steelhead. Several findings and recommendations have stemmed from this 2015 assessment:

- Pearson et al. (2015) recommended additional diet research on double-crested and Brandt's cormorants and Caspian terns to determine if they consume steelhead. Because their 2015 assessment was focused on marine predation but not freshwater predation, a different suite of potential avian predators may have been identified in the freshwater systems of the Puget Sound region. Although they did not find evidence that any species has a population-level effect on salmon, their synthesis does describe several physical scenarios and/or categories of birds that may deserve further study or observation.
- Based on the 2015 assessment, WDFW began collecting data on double-crested cormorant diet at the Woodard Bay nesting colony in South Puget Sound, which coincided with management to reduce nesting in the lower Columbia River. Preliminary results indicate that salmon were a minor component of the nesting cormorant diet at the site (WDFW, unpublished data). Understanding potential population-level effects of cormorant predation or potential benefits of management would require a more comprehensive assessment of year-round diet.

- In Puget Sound, Pearson et al. (2015) contend the prospects for salmon predation by gulls appeared limited because the most abundant large gulls in Puget Sound are not able to pursue prey far under the surface. While that may be true in marine waters, there may be potential for glaucous-winged gulls (common in Puget Sound throughout the year) and the summer influx of young and post-breeding California gulls to prey on outbound salmon at the mouths of shallow creeks and river mouths as has been documented in California (Frechette et al. 2012)—however this phenomenon has not been studied or documented in the Salish Sea.
- Because of the paucity of other forage fish like surf smelt and herring in the southern portions of Puget Sound, juvenile salmon may be the primary forage fish available to predators in that region. Pearson et al. (2015) contend that: “As a consequence, we might expect an increase in juvenile salmon predation risk with decreasing latitude in Puget Sound.” However, since the time of this study, anchovy have become more prevalent in Puget Sound, which may mitigate this risk. The dynamic nature of forage fish population demographics and seasonal changes points to the importance of using up to date information to evaluate this possibility. The subset of avian predators and conditions outlined here (e.g., cormorants in winter/spring, common mergansers, gull aggregations in shallow creek/river mouths) might be studied in areas like the Nisqually River Delta to better evaluate potential effects on salmon.

Pearson et al. (2015) did not assess the diet of the common merganser, a piscivorous species more abundant in freshwater environments. To our knowledge, there are no studies on diet composition in mergansers in Puget Sound. The species occupies freshwater rivers during salmon runs, and it is likely the only species of merganser that could have an impact on juvenile salmonid survival.

Coastal Washington

There is little documentation of avian predation impacts to salmon recovery on the coast of Washington. One recent study did find strong spatial overlap between common murre, sooty shearwaters, and juvenile Chinook salmon, suggesting the possibility of predation early in the marine life stage of the fish (Zamon et al. 2024). While the authors acknowledge a low likelihood of salmonids being a majority of prey consumed by either species, they cite evidence of juvenile salmon in murre diets and suggest further specific investigation of predation in this system. One starting point for examining this relationship could involve an analysis of the California Current Predator Diet Database (Szoboszlai et al. 2015).

Figure 22. Table of relative estimates of factors impacting salmon recovery, excerpted from the Phase 2 Report (2020) of the Columbia Basin Partnership Task Force.

FIGURE 13. Heat map of impacts of limiting factors by stock and region, including ranges reflecting uncertainties where appropriate. Units are percentage reductions in equilibrium abundance (generally equivalent to mortality rates).

	Stock	Tributary Habitat	Estuary Habitat	Hydro (mainstem)	Hydro (latent)	Hydro (blocked)	Predation	Fishery	Hatchery
Lower Columbia	Spr Chinook	85	17	0	0 (0-0)	30	14	17	29 (4-54)
	Fall (tule) Chinook	70	21	0	0 (0-0)	15	11	33	25 (3-47)
	Fall (bright) Chinook	10	21	0	0 (0-0)	40	11	47	0 (0-0)
	Chum	95	50	5	0 (0-0)	0	2	1	10 (1-18)
	Coho	80	11	0	0 (0-0)	5	13	17	22 (3-42)
	Sumr Steelhead	65	28	4	0 (0-0)	40	19	5	8 (1-15)
	Win Steelhead SWW	60	28	0	0 (0-0)	0	19	5	17 (2-33)
	Win Steelhead LCR	65	28	0	0 (0-0)	10	19	5	9 (1-16)
Willamette	Spr Chinook	85	20	0	0 (0-0)	50	19	13	25 (3-46)
	Win Steelhead	80	28	0	0 (0-0)	20	32	3	2 (0-4)
Middle Columbia	Spr Chinook	85	17	23	14 (3-25)	25	25	15	24 (3-45)
	Fall Chinook	20	27	13	9 (2-17)	5	10	55	0 (0-0)
	Coho	NA	11	30	19 (5-33)	0	17	22	NA
	Sockeye	0	17	19	9 (2-17)	95	8	3	NA
	Sumr Steelhead	80	28	11	14 (3-25)	20	33	10	17 (2-33)
Upper Columbia	Spr Chinook	45	18	49	38 (9-67)	75	29	15	32 (5-59)
	Summer Chinook	50	27	49	38 (9-67)	50	13	61	27 (4-51)
	Fall Chinook	25	27	65	19 (5-33)	5	13	61	10 (1-18)
	Sockeye	50	17	38	38 (9-67)	80	24	12	10 (1-18)
	Sumr Steelhead	40	31	30	38 (9-67)	95	52	10	24 (3-45)
Snake	Spr Chinook	50	16	39	38 (9-67)	30	29	14	15 (2-28)
	Fall Chinook	25	27	62	38 (9-67)	80	13	45	NA
	Sockeye	10	17	47	38 (9-67)	70	24	6	NA
	Sumr Steelhead	45	27	30	38 (9-67)	40	43	25	24 (3-45)
<5%		5-20%		21-30%		31-50%		>50%	

Complexities and knowledge gaps

Complexities

Understanding predator-prey relationships

Food webs are intrinsically complicated, and changes in predator populations can alter the structure and dynamics of entire ecosystems. Understanding interactions between species is crucial for conservation efforts, as management can have unanticipated consequences.

Interconnected species threats

There are multiple, often interacting threats to both salmon and birds (as described in this report's "State of current knowledge" section). It is extremely difficult and often impractical to understand and consider a single threat in isolation from others.

Considering modified environments

Determining management strategies within highly modified environments such as the Columbia River System is complicated, as the desired baseline is often some undefinable "natural state."

For example, the Columbia River System has been substantially altered by human activities, including development for hydropower infrastructure. There are 14 federally owned and operated projects, including hydropower dams, and five public utility dams on the mainstem Columbia and lower Snake Rivers in Washington. In addition, there are several other dams on the Yakima River, upper Snake River, and other Columbia River tributaries. These structures have been in place for 50 to nearly 100 years, changing salmon habitat and movements, raising water temperatures, altering flow patterns, and affecting the movement of sediment and contaminants. Most, if not all, of these changes have been detrimental to salmon. Other salmon recovery regions in Washington, including Puget Sound, generally have less dramatic manipulation of salmon habitat but inevitably, human activities have still altered the landscape – typically not to salmon's benefit.

On the Columbia River, one impact of human use is dredging and the subsequent deposition of the dredged materials, often in the form of in-river islands. These islands provide suitable habitat for colonial ground-nesting piscivorous birds such as terns, pelicans, and cormorants. Many historic nesting sites are unsuitable due to human encroachment and/or decreased water levels (Suryan et al. 2004). At the same time, hatchery production has increased the availability and concentration of juvenile salmon in rivers. The combination of predator-free islands in the Columbia River and the increase in prey availability create favorable nesting conditions for birds that might otherwise have nested elsewhere.

Measuring impacts and effects

Recovery actions need to be grounded in scientific evidence (see "Intervention Principles," this document). However, due to the complexity inherent in the migratory life cycle of salmon, it is nearly

impossible to quantify the impact of any factor in isolation. Similarly, measuring the impact of a given management action in an ecosystem is challenging. The primary goal of any management for salmon is to encourage returning adults that spawn and perpetuate populations. While measurable, variation in those adult returns is driven by a multitude of factors between egg and return: the juvenile journey to the ocean, ocean stage, and then the adult run back to the spawning river.

Distinguishing between additive vs. compensatory mortality

It is difficult but important to try to understand whether instances of avian predation lead to additive mortality of juvenile salmonids, compensatory mortality, or a combination of the two. Additive mortality indicates the birds are consuming fish that would have otherwise lived to spawn, whereas compensatory mortality indicates the fish would have died at some point before spawning regardless of avian predation. A general assumption is if a bird consumes a juvenile salmonid that would otherwise die before spawning (compensatory mortality), then its predation would not contribute to an impact on salmonid recovery at a population level; but if a bird consumes a juvenile salmonid that would have otherwise lived to spawn (additive mortality), its predation could contribute to an impact on salmonid recovery at a population level.

Knowledge gaps

Limited avian diet information

It is difficult to quantify diet and foraging dynamics for species that are flexible and adaptive in their prey requirements and have highly migratory behavior and large geographic ranges that span international, state, tribal, and local jurisdictions. Current monitoring efforts, which include nest counts, productivity surveys, and mark-recapture analyses, are not state-wide. The monitoring data can inform targeted management actions but does not necessarily provide enough information to fully understand specific management implications.

Asymmetrical regional data and knowledge

Relatively little is known about avian predation on salmonids in Washington outside the Columbia River System. The effects of avian predation on juvenile salmonids have not been as well studied on the coast or in the Salish Sea and associated riverine systems; this is described in this report's "State of current knowledge" section.

Lack of clear management goals

Avian predation management plans should be implemented with a clear goal regarding the anticipated benefit to salmonids in mind. Returning adult fish abundance, widely considered the "gold standard" for monitoring salmon, cannot be attributed to any particular condition or action because of the many factors that influence survival during the salmon life cycle. Therefore, establishing goals and metrics that are meaningful and achievable is a challenge.

ASPWG findings

Working group decisions and acknowledgements

The Legislature's requests for more information about avian salmon predation are far from simple. As the Working Group grappled with the inherent complexities and knowledge gaps described previously, they made three key decisions and several acknowledgements to provide responses in this report:

Decisions

- The Working Group sharpened the focus of their discussions on juvenile salmonid survival (rather than salmonid recovery in general) of both wild- and hatchery-origin ESA-listed stocks.
 - Salmonids are most vulnerable to avian predation at the beginning of their life cycle when they leave the freshwater system and migrate to the ocean. In the marine environment and at later life cycle phases, the fish are exposed to several other mortality factors. Given this, the Working Group considered avian predation management as a targeted intervention to increase the number of juvenile salmonids that have the opportunity to reach the ocean.
- The Working Group expanded the scope of their discussions to include avian salmon predation intervention principles.
 - The Working Group's collective expertise allowed them to consider and describe avian management approaches that have been used, are currently in place, and/or may be considered (see Appendix D), which sets them up well to provide additional guidance.
- The Working Group did not identify a need for new or revised state statutes and regulations related to avian predation management at this time.

Acknowledgements

The Working Group's discussions reflected the following acknowledgements, which also underpin their responses to the Legislature in this report:

- The Working Group recognizes the value of both piscivorous birds and salmonids—both predators and prey—in the region.
- The Working Group recognizes no management action exists in a vacuum; management actions should be designed, implemented, and adapted with a holistic approach that acknowledges the nuances of the specific watershed, salmon recovery efforts within it, and avian behaviors and population statuses.
- The Working Group acknowledges the definitions of “population level” and “impact” are fluid; different working definitions may be appropriate for different management needs.
- The Working Group recognizes that many of the state's environments where birds prey on fish are highly modified ecosystems that have significantly influenced both bird behavior and salmonid survival.

- The Working Group believes multi-benefit management actions are possible and highly preferable.

Focal avian species

Legislative request 1: Identify all avian species that contribute to predation of juvenile salmon at a population level.

While there are many bird species that consume juvenile salmon in Washington (see Appendix C, “Table of Washington piscivorous birds”), four primary species or species groups have life history characteristics that may result in negative impacts to juvenile salmon at a population level: American white pelicans, Caspian terns, double-crested cormorants, and gulls. Additionally, common merganser predation of salmonids merits more study, particularly where they overlap ESA-listed fish stocks. These bird species are a concern because their diet is primarily fish; they are present in Washington waters when salmon typically run in the spring and fall (especially in upstream or freshwater locations, i.e., not estuaries, where there is more prey variety); they nest colonially, concentrating predation around the nesting site; and/or juvenile salmon are susceptible to their foraging techniques, such as diving or coordinated hunting.

Avian predation impacts on salmonids

Legislative request #2: Determine whether such species are adversely impacting the recovery of any threatened or endangered salmon species.

Determining whether instances of avian predation impact juvenile salmon survival must be done on a case-by-case basis; it is impossible to make a general statement such as “Bird species [X] impacts threatened or endangered salmon species [X].”

For each instance of predation, a clear predation threshold should be developed using watershed- and salmon stock-specific predation rates and adult returns; an understanding of how predation impacts treaty rights and obligations; and an understanding of economic implications of the management actions, such as the cost-effectiveness of monitoring.³

The ASPWG adopted language developed by the Pacific Flyway Council for addressing migratory bird predation issues on fish resources (Pacific Flyway Council Policy Statement—Avian Predation on Fish Resources, 2011): “Responses to perceived avian predation issues [should be] based on sound science.” Specifically:

- A. Magnitude and scope of predation impacts are best demonstrated through scientific evidence.⁴

³ “Predation threshold” refers to the amount of threatened or endangered juvenile salmon a bird species can eat in a watershed or area before it impacts that stock’s ability to recover.

⁴ The ASPWG believes “science” is a more appropriate term than “empirical evidence,” which is the language used by the Pacific Flyway Council.

- B. Monitoring, data sharing, data gaps, and research needs are acknowledged and addressed.
- C. Expectations of how management actions will reduce impacts to affected fish populations are explicitly addressed.
- D. Expected outcomes of management actions on affected avian populations are clearly understood
- E. Measures are implemented to assess effectiveness of management actions and inform future direction (i.e., adaptive management).

Additionally:

- Predation thresholds are not defined for many threatened and endangered salmon populations or watersheds.
- It is difficult to establish a “gold standard” to assess predation or efficacy of management. For example, high-quality monitoring plans are often impractical to implement due to high costs and complex environments. Considering this challenge, thoughtful scientific approaches are needed to support avian management.
- Predation impacts largely depend on the extent to which a predator and its prey overlap in time and space. Juvenile salmon survival is population-specific and depends on vulnerability associated with the prey life stage and environmental conditions that vary across years. The specific risk of mortality due to avian predation also depends on the migration route relative to established bird colonies and water conditions (e.g., velocity, turbidity, etc.).
- Different management actions require different levels of evidence. For example, local dissuasion or non-lethal tools may require less biological justification and monitoring than large-scale and/or lethal management actions.

Possible remedies

Legislative request #3: Identify remedies.

The ASPWG does not have the scope or resources to articulate clear, stand-alone avian predation problem statements and suggest specific remedies for specific areas in the state. However, it developed two products to respond to the Legislature’s request:

1. Intervention Principles that guide the development of avian predation management plans and remedy approaches.
2. With support from the Department, the ASPWG developed an inventory of remedies that have been implemented or are being implemented in the region (Table 2). The remedy inventory table outlines the general category of intervention method, specific intervention method, whether or not the intervention requires a Migratory Bird Treaty Act permit from USFWS, and regional examples of the intervention in use (if applicable). See Appendix E, “Examples of remedy outcomes” for more outcomes associated with the regional examples.

ASPWG Intervention Principles

If avian predation is determined to adversely impact juvenile salmonid survival (as discussed previously), actions to address it should be appropriately designed. In each instance, there are scientific, cultural, economic, and treaty elements that must be considered. The ASPWG developed the following five intervention principles to support the development of robust, long-term adaptive management plans for reducing avian predation, increasing juvenile salmonid survival, and building resiliency into salmon populations.

An adaptive management plan should include remedies that are:

1. Effective.
2. Consistent with laws and regulations.
3. Aligned with ecosystem-based principles.
4. Cognizant of a diversity of cultural values.
5. Actionable.

1. The remedy is effective if it:

- a. Addresses a known problem in a specific action area.
- b. Has plans in place to define metrics and evaluate success.
- c. Increases the number and, if possible to determine, the quality of juvenile salmonids leaving the action area.
- d. Reduces (or eliminates) avian predation to target levels in a defined area, without displacing the effects to another area where it might jeopardize salmon recovery.

2. The remedy is consistent with laws and regulations if it:

- a. Complies with federal regulations under the Migratory Bird Treaty Act and the Endangered Species Act.
- b. Complies with treaty-reserved rights and stewardship needs for trust resources.
- c. Complies with state laws and regulations.

3. The remedy is aligned with ecosystem-based principles if it:

- a. Increases juvenile salmonid survival while reducing take of avian predators (balances the long-term population of each species while considering individual species population objectives).
- b. Recognizes predation in context, particularly that it frequently occurs in highly altered systems.
- c. Acknowledges lessons learned and avoids unintended consequences to local and distant areas and ecosystems.
- d. Recognizes certain colonies and/or predation pinch points (i.e., hotspots) are worth focusing on and simultaneously acknowledges the scale of specific problems is much smaller than the scale of broader ecosystem recovery needs.
- e. Is based on best available science (may need to be applied from other areas).
 - i. Bird species specific (what works for one species doesn't work for another).

- ii. Fish species-specific (stock type, stream type, and ocean type), e.g., steelhead are surface-oriented as juveniles and can be more vulnerable to predation.

4. The remedy recognizes a diversity of cultural values if it:

- a. Acknowledges there is a variety of cultural values involved, including animal rights.
- b. Acknowledges environmental justice implications, including negative impacts on coastal and tribal communities.
- c. Remedy proponents communicate concepts *a* and *b* to the public and interest holders.

5. The remedy is actionable if it is:

- a. Collaborative and implemented in conjunction with regional cooperation and/or other recovery efforts and managers.
- b. Cost-effective.
- c. Safe to implement.

Remedy inventory

Table 2 lists actions to dissuade avian predation; it does not include actions that solely involve salmon or salmon habitat (e.g., actions to improve juvenile salmon rearing habitat or increasing alternate prey for birds). When considering remedies, the ASPWG emphasizes the following:

- Any avian predation remedy should align with the ASPWG Intervention Principles.
- Managers should do their best to consider possible population-level impacts on birds and avian life history needs when dissuading nesting birds. One technique would involve “push-pull,” in which managers “push” birds from one location and “pull” them to a neighboring area via attractants where their impacts on juvenile salmonids are expected to be reduced.
- Remedies are rarely used in isolation. In practice, multiple remedies are used in conjunction with one another – and often different combinations are attempted – to achieve the desired results.

Table 2. Potential avian predation management actions, including examples in Washington and/or the Columbia River.

This list is intended to be illustrative, not exhaustive, and does not include activities to increase prey options for piscivorous birds or management focused on improvement of other aspects of salmon habitat.

Action (general)	Action (specific)	MBTA Permit required*	WA example(s)
Removal	Trapping/relocating	yes	
Removal	Egg/nest take	yes	
Removal	Adult lethal take	yes	
Hazing	Falconry abatement	no	Hanford Reach, Miller Rocks Island
Hazing	Human presence	no	East Sand Island

Action (general)	Action (specific)	MBTA Permit required*	WA example(s)
Hazing	Dog presence	no	East Sand Island
Hazing	Noise (e.g., horns, cracker shells)	no	Bonneville Dam
Hazing	Laser	no	Crescent Island
Habitat manipulation	Dissuasion (flags, fencing, etc.)	no	East Sand Island, Badger Island
Habitat manipulation	Spikes	no	Rice Island
Habitat manipulation	Decoys (predators)	no	Miller Rocks Island
Habitat manipulation	Effigies	no	Miller Rocks Island
Habitat manipulation	Flooding	no	Blalock Islands
Habitat manipulation	Changing vegetation	no	Badger Island, Troutdale Towers

*A Migratory Bird Treaty Act (MBTA) permit is required for the purposeful take of adult birds and active nests including eggs or chicks. The actions above the double line inherently involve nest or bird removal, while the actions below the double line would require a permit if applied to active nests or nesting individuals.

ASPWG recommendations

1. Fund research to examine geographically specific anecdotes of salmon predation by birds, particularly Puget Sound rivers with at-risk salmon stocks and that are cited as places where avian salmon predation negatively impacts juvenile salmonid survival, including but not limited to the Stillaguamish River and the Nisqually River Delta.
2. Invest in WDFW's capacity to coordinate with federal and state governments, tribes, and other regional entities to address avian predation issues in the Columbia River System.
3. Invest in Department-led work focused on closing data gaps and learning more about the shared needs of birds and salmon. This work can make critical connections between ecosystem recovery actions that benefit both species, increase the Department's ability to partner in avian predation policy and science, and promote coordination with tribes and other partners. For example:
 - a. Update Washington seabird colony inventory and diet database.
 - b. Research piscivorous bird species diet and foraging dynamics in Washington state.

Appendix A. ASPWG membership

Table 3. Avian Salmon Predation Working Group members.

ASPWG member name	Title and affiliation
Aaron Brooks	Fisheries Management Biologist, Jamestown S’Klallam Tribe
Bill Sharp	Klickitat Coordinator, Yakama Nation
Bryce Divine	Commercial fisheries
Chris Magel	Natural Resource Management Specialist, National Marine Fisheries Service – West Coast Region
Clark Watry	Project Leader, Nez Perce Tribe
David Troutt	Natural Resources Director, Nisqually Tribe
Emma Sands	Harvest Management Biologist, Quileute Tribe
James Lawonn	Avian Predation Coordinator, Oregon Department of Fish and Wildlife
Jennifer Urmston	Permits Biologist, U.S. Fish and Wildlife Service
Jessica Stocking	Marine-Coastal-Flyway Section Manager, Washington Department of Fish and Wildlife
Joy Lee Waltermire	Senior Fish Biologist, Long Live the Kings
Larry Phillips	Pacific Fisheries Policy Director, American Sportfishing Association
Robert Sudar	Commercial fisheries
Ron Garner	President, Puget Sound Anglers
Sean Tackley	Fish Policy and Program Manager, U.S. Army Corps of Engineers – Northwestern Division
Todd Hass	Special Assistant to the Director, Puget Sound Partnership
Trina Bayard	Director of Bird Conservation, Audubon Washington

Appendix B. ASPWG process overview

In summer 2024, the Washington Department of Fish and Wildlife (WDFW, the Department) convened the Avian Salmon Predation Working Group (ASPWG, Working Group) in response to SHB 2293 (described in “Report Purpose”), with the first meeting taking place in November.

The ASPWG met five times to discuss avian salmon predation and develop this report in response to the Legislature’s request. Meeting agendas, presentations, and notes are available on the Department’s [ASPWG webpage](#). Working Group members routinely reviewed materials and provided feedback to the Department’s facilitation team between meetings.

1. **Meeting 1: November 7, 2024.** The ASPWG kickoff meeting was dedicated to introducing members of the Working Group to one another and reviewing the Working Group’s legislative charge. The Department provided an orientation to the challenge of considering and managing avian salmon predation and the group discussed the scope of the Working Group’s task.
2. **Meeting 2: February 4, 2025.** The second ASPWG meeting was dedicated to building shared awareness and understanding between Working Group members about various topics related to avian salmon predation. The meeting included presentations on Washington salmon recovery and piscivorous birds by WDFW subject matter experts, and a presentation about federal regulatory frameworks by the U.S. Fish and Wildlife Service.
3. **Meeting 3: March 17, 2025.** The ASPWG discussed how to develop guidance for managers in situations where avian predation causes negative impacts to juvenile salmonid survival. This discussion bridged a gap between the Legislature’s second and third requests about identifying impacts of avian predation and possible remedies. The ASPWG also reviewed a draft outline for its final report.
4. **Meeting 4: May 6, 2025.** The ASPWG continued its conversation about guidance to develop a list of Intervention Principles and discussed its approach to identify remedies to avian salmon predation.
5. **Meeting 5: June 3, 2025.** Group members finalized sections of the Working Group’s report, including the Intervention Principles, the inventory of possible remedies, and its ultimate recommendations to the Legislature.

Appendix C. Table of Washington piscivorous bird species

Table 4. Washington piscivorous bird species.

Species	Statewide Trend*	Time present	Peak % of pop.	Waters inhabited	Diet	Conservation status **
Pelagic cormorant	-5.5%	breeding	2.45%	marine	fish only	none
Double-crested cormorant	-5.4%	year-round	1.7%	both	fish only	none
Brandt's cormorant	Unk.	year-round	Unk.	marine	fish only	None, BCC
American white pelican	+8.3%	breeding	1.2%	freshwater, estuarine	fish	State sensitive, BCC
Brown pelican	-38%	post-breeding	2.3%	marine-outer coast only	marine fish	SGCN
Common loon	-13.7%	year-round	2.9%	Lakes, marine	primarily fish	State sensitive, SGCN
Pacific loon	Unk.	non-breeding	10.9%	marine	primarily fish	none
Red-throated loon	Unk.	winter	8.7%	marine	fish only	none
Common merganser	-1.76%	year-round	1.4%	Freshwater-larger rivers, lakes	primarily fish	none
Red-breasted merganser	-0.68%	winter	2.4%	marine	majority fish	none
Hooded merganser	-5.76%	year-round	3.7%	freshwater	44% fish	none
Western grebe	-11%	year-round	17.5%	both	80-100% fish	SGCN, BCC
Clark's grebe	Unk.	Year-round	0.84%	both	no data	SGCN, BCC
Red-necked grebe	17.3%	winter	13.8%	both	~55% fish	SGCN

Species	Statewide Trend*	Time present	Peak % of pop.	Waters inhabited	Diet	Conservation status **
Horned grebe	-28.6%	winter	15.2%	marine	fish, invertebrates	none
Pied-billed grebe	-2.46%	year-round	2.2%	freshwater	generalist	none
Common murre	Unk.	year-round	Unk.	marine	fish, invertebrates	none
Rhinoceros auklet	Unk.	year-round	Unk.	marine	fish, invertebrates	none
Pigeon guillemot	-23.7%	year-round	59.4%	marine	fish, invertebrates	none
Marbled murrelet	-23.7%	year-round	1.2%	Marine, occasionally freshwater	fish	Federally threatened, state endangered, SGCN
Caspian tern	+6.6%	breeding	15.9%	both	fish	none
Forster's tern	+2.4%	breeding	0.9%	freshwater	fish	None, BCC
California gull	-21.3%	year-round	12.4%	both	generalist	None, BCC
Ring-billed gull	-17.5%	year-round	1.1%	freshwater	generalist	none
Western gull	-22.8%	year-round	2.2%	Marine (outer coast)	generalist	None, BCC
"Olympic" gull	-33.4%	year-round	3.8%	Marine (sound)	generalist	none
Belted kingfisher	-18%	breeding	1.7%	both	primarily fish	none
Osprey	+0.16%	breeding	5.2%	both	fish	none
Great blue heron	-4.1%	breeding	1.4%	freshwater	generalist	none

Species	Statewide Trend*	Time present	Peak % of pop.	Waters inhabited	Diet	Conservation status **
Black-crowned night heron	-26.2%	year-round	0.02%	freshwater	generalist	none
Great egret	-12.3%	year-round	0.01%	freshwater	generalist	none

*Statewide trend is for the years 2011-2021, which does not capture more recent changes in bird distribution or abundance in Washington due to management actions or avian influenza. These trends are based on population modeling derived from community-science eBird data.

**Species of Greatest Conservation Need (SGCN) are identified in Washington's 2015 State Wildlife Action Plan; an update to the Plan and the list of SGCN species will be published in late 2025. Birds of Conservation Concern (BCC) are identified by U.S. Fish and Wildlife Service.

Appendix D. Avian salmon predation in Washington

Table 5 highlights areas of concern for potential avian predation impacts to salmon.

Region of Origin: Recovery region as defined by the Governor's [State of the Salmon](#).

ESU/DPS - Evolutionarily Significant Unit/Distinct Population Segment: These are the functional units at which salmonids are listed as Threatened or Endangered under the Endangered Species Act. There is one Endangered DPS, Upper Columbia Spring Run Chinook.

Status: Federal listing status for the ESU/DPS.

Populations: The number of unique populations identified in a given DPS. We have 151 populations in Washington.

Areas of potential avian predation concern: This narrows the geography as identified by areas of potential concentrations of avian predation as defined by WDFW staff or in recovery plans for the salmon DPSs.

Primary period of vulnerability: The time window when salmon smolts move through the areas of potential predation concern.

Pinnipeds present: Whether seals and/or sea lions (pinnipeds) are present in the area of potential concern. Pinnipeds are often identified in salmon recovery plans as potentially impeding recovery of DPSs that encounter them at sensitive life stages or concentrations.

Fish predators present: Which potential fish predators are identified in the Recovery Plan for each ESU/DPS.

Bird colonies present: Bird species identified by NOAA Fisheries (NMFS) to be hindering listed salmon recovery in some areas and nesting in the Area of Potential Concern.

Avian predation estimates: Estimates of avian predation derived primarily through studies of deposition rates of passive integrated transponder (PIT) tagged, hatchery-raised fish at bird nesting colonies (Roby et al. 2021; Evans et al. 2024). AWPE = American white pelican; CATE = Caspian tern; DCCO = double-crested cormorant; GULL = gull species.

Table 5. Areas of concern for potential avian predation impacts to salmon in Washington.

Region of origin	ESU/DPS	Status	Populations	Areas of potential avian predation concern	Primary period of vulnerability	Pinnipeds	Fish predators	Bird colonies	Avian predation estimates
Hood Canal	Hood Canal Summer Run Chum	Threatened	2	In Puget Sound, Salish Sea	March-April	Y	other salmonids (esp. at hatchery releases); predation from seabirds, marine fish and marine mammals is generally low	CATE, DCCO, GULL	none
Puget Sound	Puget Sound Chinook	Threatened	22	In Puget Sound, Salish Sea	March - April	Y	n/a	CATE, DCCO, GULL	none
Puget Sound	Puget Sound Steelhead	Threatened	32	In Puget Sound, Salish Sea	March - June	Y	n/a	CATE, DCCO, GULL	none
WA Coast	Lake Ozette Sockeye	Threatened	1	Lake Ozette watershed	April?	Y	cutthroat trout, northern pikeminnow, largemouth bass	none	none, likely not a primary concern
Lower Columbia	Columbia River Chum	Threatened	11	Unknown, estuary colonies?	?	Y	salmon smolts, pikeminnow, walleye, smallmouth bass	CATE, DCCO, GULL	none
Lower Columbia	Lower Columbia River Chinook	Threatened	22	Columbia River Estuary (ESI and AMB)	March - June	Y	northern pikeminnow, walleye and smallmouth bass	CATE, DCCO, GULL	CATE: 4.1% before management, 2.5% during management; DCCO: 27.5% before management, 1-7% during management; gulls: <i>no info</i>
Lower Columbia	Lower Columbia River Coho	Threatened	17	Columbia River Estuary (ESI and AMB)	April - June	Y	northern pikeminnow, walleye and smallmouth bass	CATE, DCCO, GULL	CATE: 2.6% before management, 3.1% during management; DCCO: 8-18% before management, .2-.3% during management; gulls: <i>no info</i>

Region of origin	ESU/DPS	Status	Populations	Areas of potential avian predation concern	Primary period of vulnerability	Pinnipeds	Fish predators	Bird colonies	Avian predation estimates
Lower Columbia	Lower Columbia River Steelhead	Threatened	19	Columbia River Estuary (ESI and AMB)	April - June	Y	northern pikeminnow, walleye and smallmouth bass	CATE, DCCO, GULL	CATE: 15.2% before management, 10.4% during management; DCCO: 2.6-5.7% before management, 1.7-11.2% during management, 0.6% after colony collapse (2017-18); gulls: <i>no info</i>
Middle Columbia	Middle Columbia Steelhead	Threatened	10	Rice Island	April - June	Y	primarily northern pikeminnow, smallmouth bass, walleye	CATE, GULL	CATE: ≤2.9% (2022-23)
Middle Columbia	Middle Columbia Steelhead	Threatened	10	Astoria-Megler Bridge	April - June	Y	primarily northern pikeminnow, smallmouth bass, walleye	DCCO	DCCO: 6.8% (2023)
Upper Columbia	Upper Columbia River Spring Run Chinook	Endangered	3	upstream of Bonneville Dam, CPR (including Foundation Island)	mid-March through mid-May	N	northern pikeminnow, walleye, and smallmouth bass most likely; trouts most likely among salmonids	AWPE, CATE, DCCO, GULL	CATE: 0.05-0.29%; DCCO: <0.01
Upper Columbia	Upper Columbia River Steelhead	Threatened	4	from middle Columbia to Pacific Ocean	April - June	N	northern pikeminnow, walleye, and smallmouth bass most likely; the trouts most likely among salmonids	AWPE, CATE, DCCO, GULL	all avian species combined: 28.6%; DCCO: 5.1%
Upper Columbia	Upper Columbia River Steelhead	Threatened	4	Columbia River Estuary (ESI and AMB)	April - June	Y	northern pikeminnow, walleye, and smallmouth bass most likely; the trouts most likely among salmonids	CATE, DCCO	DCCO: variable, from 0.7-11.4%, 9.2% at AMB (2022-23)

Region of origin	ESU/DPS	Status	Populations	Areas of potential avian predation concern	Primary period of vulnerability	Pinnipeds	Fish predators	Bird colonies	Avian predation estimates
Upper Columbia	Upper Columbia River Steelhead	Threatened	4	Rice Island	April - June	Y	northern pikeminnow, walleye, and smallmouth bass most likely; the trouts most likely among salmonids	CATE, GULL	CATE: ≤1.1% (2022-23)
Upper Columbia	Upper Columbia River Steelhead	Threatened	4	upstream of Bonneville Dam, CPR	April - June	N	northern pikeminnow, walleye, and smallmouth bass most likely; the trouts most likely among salmonids	AWPE, CATE, DCCO, GULL	CATE: during management 1.0%; DCCO: <0.01 (higher between Rock Island and McNary)
Upper Columbia	Upper Columbia River Steelhead	Threatened	4	McNary Dam area	April - June	N	northern pikeminnow, walleye, and smallmouth bass most likely; the trouts most likely among salmonids	AWPE, CATE, DCCO, GULL	CATE: 1.4% (2023)
Snake River	Snake River Basin Steelhead	Threatened	4	from lower Snake R to Pacific Ocean	April - June	Y	pikeminnow, smallmouth bass, walleye, brook trout	AWPE, CATE, DCCO, GULL	cumulative, all avian species estimate: 23.2%
Snake River	Snake River Basin Steelhead	Threatened	4	Foraging distance from Foundation and Badger Islands in McNary Reservoir	April - June	N	pikeminnow, smallmouth bass, walleye, brook trout	AWPE, CATE, DCCO, GULL	CATE: 1.9% (Badger Island, 2023); DCCO: 3.1%
Snake River	Snake River Basin Steelhead	Threatened	4	Astoria-Megler Bridge	April - June	Y	pikeminnow, smallmouth bass, walleye, brook trout	DCCO	DCCO: 10.9% (2023)
Snake River	Snake River Fall Run Chinook	Threatened	1	Astoria-Megler Bridge	late April - late June	Y	pikeminnow, smallmouth bass, walleye, channel catfish	DCCO	DCCO: 3.8% (2023)

Region of origin	ESU/DPS	Status	Populations	Areas of potential avian predation concern	Primary period of vulnerability	Pinnipeds	Fish predators	Bird colonies	Avian predation estimates
Snake River	Snake River Fall Run Chinook	Threatened	1	transmission towers in upper CRE and LCR	late April - late June	Y	pikeminnow, smallmouth bass, walleye, channel catfish	DCCO	DCCO: 0.5%
Snake River	Snake River Spring and Summer Run Chinook	Threatened	3	Columbia Plateau Region- specifically foraging distance from Foundation Island in McNary Reservoir	mid-March through mid-May	N	pikeminnow, smallmouth bass, walleye, brook trout	DCCO	DCCO: 2.1%
Snake River	Snake River Spring and Summer Run Chinook	Threatened	3	Astoria-Megler Bridge	mid-March through mid-May	Y	pikeminnow, smallmouth bass, walleye, brook trout	DCCO	DCCO: 7.1%

Appendix E. Examples of outcomes from “remedies” to manage avian predation

Examples of management actions to reduce avian predation of salmon, including details of attempted use and outcomes.

Columbia River Estuary

East Sand Island (ESI)

Phase I management started in earnest in 2012 (reduction of habitat, active dissuasion) Phase II management (lethal on DCCO, extreme hazing) was 2015-2019. Caspian terns nesting outside the established area were passively dissuaded using landscape features (disking, fencing, area shape changes) in 2010.

0% breeding success in 2023 by CATE, lowest level ever recorded (524 breeding pairs). Complete breeding failure in last six of eight years due to secondary predation of eggs by gulls during bald eagle predation attempts; and avian influenza, which was estimated to have caused the direct mortality of 10-14% of the Pacific Flyway population in 2023. Almost no cormorant nesting for 2019-2023.

Astoria Megler Bridge

Dramatic increase of cormorant nesting on bridge since management of ESI after 2015. 5,153 breeding pairs in 2023. Located in freshwater mixing zone. Some hazing and egg take occurs annually.

Longview/Lewis and Clark Bridge

Increase in DCCO nesting since management of ESI.

Troutdale Transmission Towers

Increase in DCCO nesting since management of ESI.

Rice Island

Rice Island is a dredge spoil island created in the 1980s, with Caspian tern management beginning in 1986. In 1999, terns nesting on Rice were relocated to East Sand Island using attractants to East Sand (decoys, audio playback, habitat restoration). Terns did not nest on the island for the next ten years but returned in 2011. Since then, active dissuasion such as stakes, fencing, hazing by people, and egg take. Active dissuasion has continued since, including in 2023, but efforts occurred after smolt outmigration period, at which time there were 1,400 terns present on Rice Island.

Mid-Columbia

Crescent Island

Passive dissuasion (woody debris placement) and planting willows began in 2015 and 2016 to deter nesting terns. Passive dissuasion materials were removed in 2020 due to success of vegetation. Despite this, the colony reestablished in 2022 and 2023 (149 and 88 breeding pairs, respectively). Placed large woody debris in 2023 pre-breeding season, moved to other areas of island. Most terns that used to nest on Crescent (prior to 2015) moved to Blalock Island in John Day.

Badger Island

Passive dissuasion (woody debris placement) used in 2022-2023, but this was still the largest tern colony in 2023.

Upper Columbia

Goose Island (Potholes)

Passive and active efforts to dissuade terns resulted in just 12 breeding pairs in 2023. Efforts included placement of fencing, rebar, and concrete blocks instead of adding vegetation due to the rocky substrate of the island. Predation rates by all terns nesting here and at Lake Lenore was less than 1%/ESU or DPS/colony. These birds presumably forage in Upper Columbia above Kennewick, Washington. Adaptive management has reduced predation rates from 15.7% prior to management (2007-2013) to 1% (0.7%-1.7%) from terns nesting at Potholes/Goose Island.

Blalock Islands

Water levels were raised at John Day reservoir per the Inland Avian Predation Management Plan, resulting in zero nesting Caspian terns in 2021-23.

Lake Lenore

Gulls, cormorants, and Caspian terns nested in 2023 and presumably forage in Upper Columbia (above Kennewick) in the Columbia Plateau Region.

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